Henry Nitzberg

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Software engineer with hands-on experience in robotics systems, ML, and full-stack development spanning ROS, C++, simulation, and production infrastructure.

Skills

Languages: C++, C, Python, JS, TS
Libraries & Frameworks: OpenCV, PyTorch, Tensorflow, NumPy, Stable Baselines, IsaacSim, ROS
Miscellaneous: Docker, CAD, Cursor, Git

Education

Tufts University GPA: 3.9 | **2021 - 2025**
B.S. Summa Cum Laude in CS, Mathematics, Studio Art

Experience

Nucleus Biologics | *Software Engineer* San Diego, CA | **July 2025 - August 2026**
NB-Lux: *End-to-end digital infrastructure platform for quoting, ordering, and operations*
- First software engineering hire; identified need to rebuild legacy platform from scratch and led adoption of a more maintainable stack (React/TypeScript + FastAPI + AWS)
- Co-own the entire codebase and infrastructure with one other engineer — responsible for architecture decisions, AWS setup (EC2, RDS, S3), and implementation
- Built integrations with QMS and inventory systems covering costing, pricing, and supply chain management
- Delivered tools for document generation/signing, chemical feasibility analysis, QC tracking, and company analytics
- Led process-defining meetings with leadership to establish workflows that will govern the product lifecycle from customer intake to delivery

CoRe Lab | *Research Assistant (Advisor: Yixin Zhu)* Peking University (remote) | **January - April 2025**
GPU Accelerated Tactile Sensor Simulation
- Contributed to a tactile sensing simulation library targeting Nvidia's Warp engine, intended for parallel RL training
- Developed Blender add-ons to streamline mesh annotation workflows for the simulation pipeline

Hayden AI | *C++ Developer Intern* San Francisco, CA | **June 2024 - September 2024**
Automatic School Bus Stop Arm Violation Detection System
- Built signal-based classifier for school bus state (stopped, driving, loading, etc.)
- Parsed UBlox GPS data into human-readable format
- Integrated video processing, GPS parsing, and state classification as dockerized nodes in a pub/sub architecture

Human Robot Interaction Lab | *Research Assistant, Robotics Engineer* Tufts University | **June 2023–May 2024**
Pathfinding and general assistant robot for hospital patients
- Built vision pipeline to detect and localize elevator doors, call panels, and floor indicators using text recognition, enabling autonomous elevator navigation
- Trained and deployed DRL agent in a custom 2D simulation with handcoded lidar sensing, and dynamic obstacles
- Developed ROS package to project RealSense 3D point cloud into 2D lidar-compatible scan, enabling full 360° coverage
- Developed DIARC component for Fetch, allowing the robot to press elevator call buttons

Projects

RL-Rover *CS138: Reinforcement Learning* | **Spring 2025**
Attempted sim-to-real transfer of a PPO-trained lidar navigation policy to a custom rover (RPi 5, tank drive, cheap lidar). Gained hands-on experience with the gap between simulation and physical hardware

Multi-LLM Classifier *CS135: Machine Learning and Data Mining* | **Spring 2025**
Won 1st place in class sentiment analysis competition, outperformed all state-of-the-art non-LLM approaches using ensemble of Ollama models.

End to End Self Driving in Simulation *CS141: Probabilistic Robotics for HRI* | **Spring 2024**
- Trained separate DCNNs for image segmentation and control (steering angle, acceleration) using custom data
- Generated, collected and augmented human driving data using CARLA
- Deployed end-to-end self-driving model in GTA V

Fan of maker projects, running, climbing, coding, smoothies, Fine Art, Netflix